

CHAPTER 5 : COMPUTER POWER SUPPLY



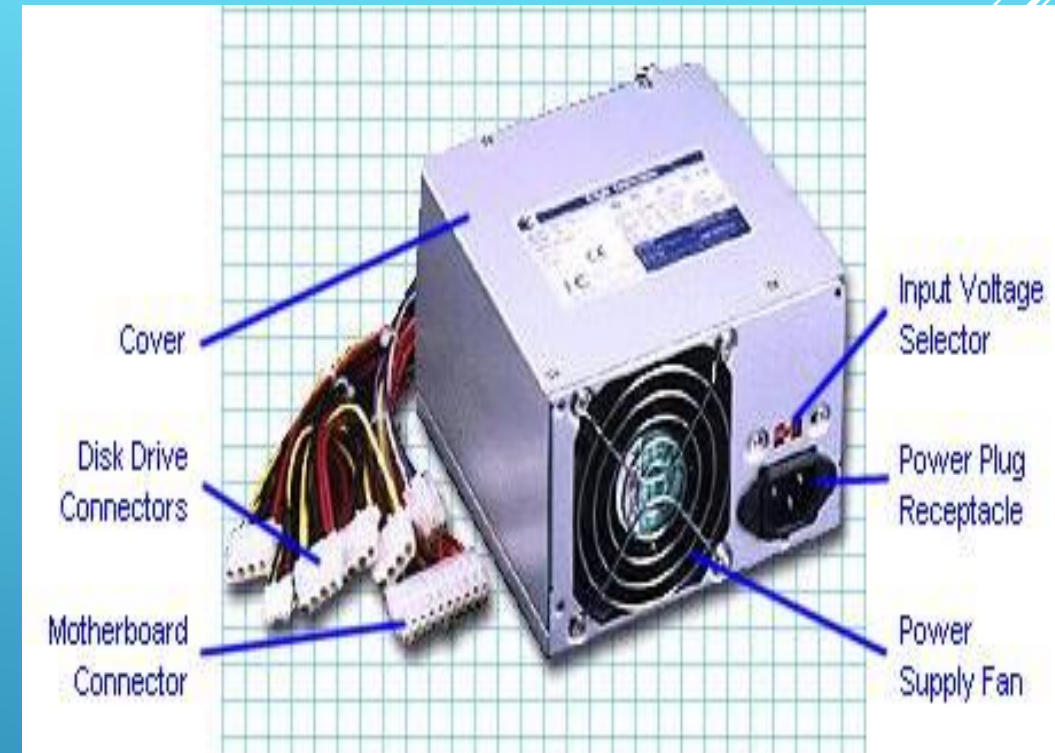
INTENDED LEARNING OUTCOMES (ILOS)

AT THE END OF THIS LESSON, THE STUDENT IS EXPECTED TO:

- ❑ BE FAMILIAR ON WHAT IS A POWER SUPPLY;
- ❑ UNDERSTAND HOW THE POWER SUPPLY WORKS;
- ❑ BE ABLE TO IDENTIFY THE PARTS AND COMPONENTS OF A POWER SUPPLY ;
- ❑ BE ABLE TO PERFORM THE BASIC TROUBLESHOOTING FOR A POWER SUPPLY UNIT ;
- ❑ ENUMERATE THE TYPE OF A POWER SUPPLY FILTERS;
- ❑ LIST ALL THE TYPES OF REGULATORS;
- ❑ UNDERSTAND THE COLOR CODING OF POWER SUPPLY CABLES;
- ❑ ENUMERATE THE DIFFERENT TYPES OF POWER SUPPLY CONNECTORS AND CABLES

PARTS OF A POWER SUPPLY

- a. Input voltage Selector – Allows you to choose which Source or input voltage you are to use (110v or 220v)
- b. Power plug receptacle – Connect the power supply to the main electricity supply via wall socket
- c. Power supply fan- Circulates air throughout the power Supply It acts as a cooling system to the power supply
- d. Cover- Simply providing structural support to the power supply and keeps dust
- e. Disk Drive Connectors- Connectors that are designed to be attached to the rear of each drive
- f. Motherboard Connector- Connects to the motherboard to provide power to the motherboard Processor, memory, chipset, integrated components and any card plugged into bus slots.



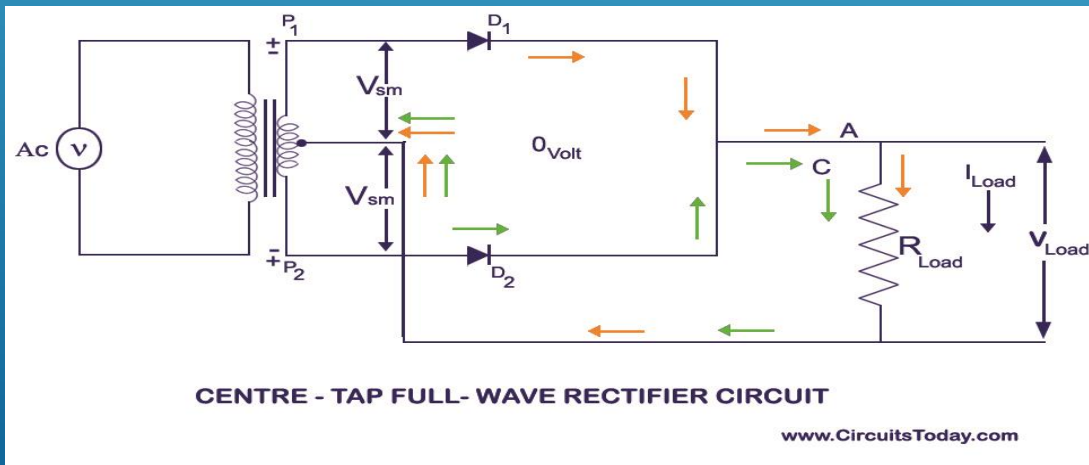
Basic Components of a Power Supply

- a. Transformer – it is a static device that transfer electrical energy from The primary winding to the secondary winding without affecting the frequency.
- b. Rectifier – it is a device used to change the ac power into dc. The basic rectifier is The diode. This diode is a unidirectional device that act as rectifier in the forward direction.

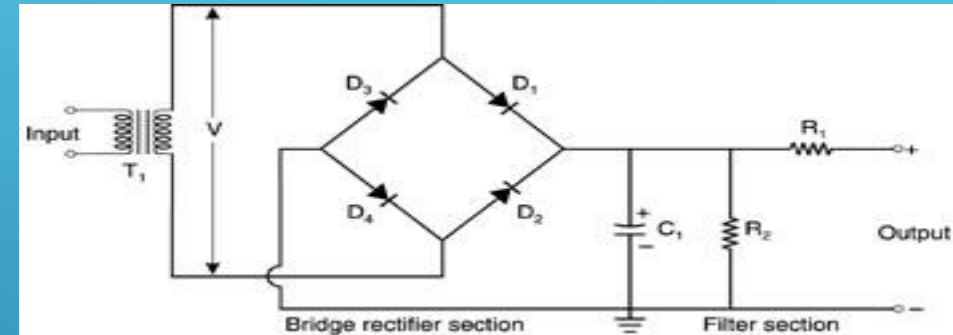
HALF WAVE



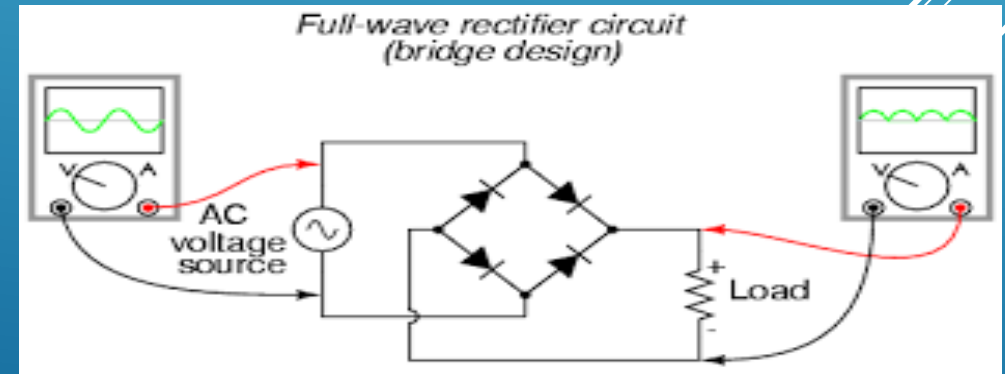
Full-Wave Center-Tapped



Full-Wave Bridge type



Full-wave rectifier circuit (Bridge Design)



2 type of Power Supply Filters

- a. C-filter or Capacitance filter – is the simplest and most economical filter available
- b. RC-filter – is used to reduce the amount of ripple voltage across a capacitor filter

Voltage Regulators

A voltage regulator is intended to give a very steady or well regulated DC output. It is always Best to have steady output voltages so that the load will function properly.

Series voltage regulator

The series element controls the amount of the unregulated input voltage that goes to The output as a regulated output.

Shunt voltage regulator

The shunt voltage regulator provides regulation by shunting current away from the load to Regulate the output voltage.

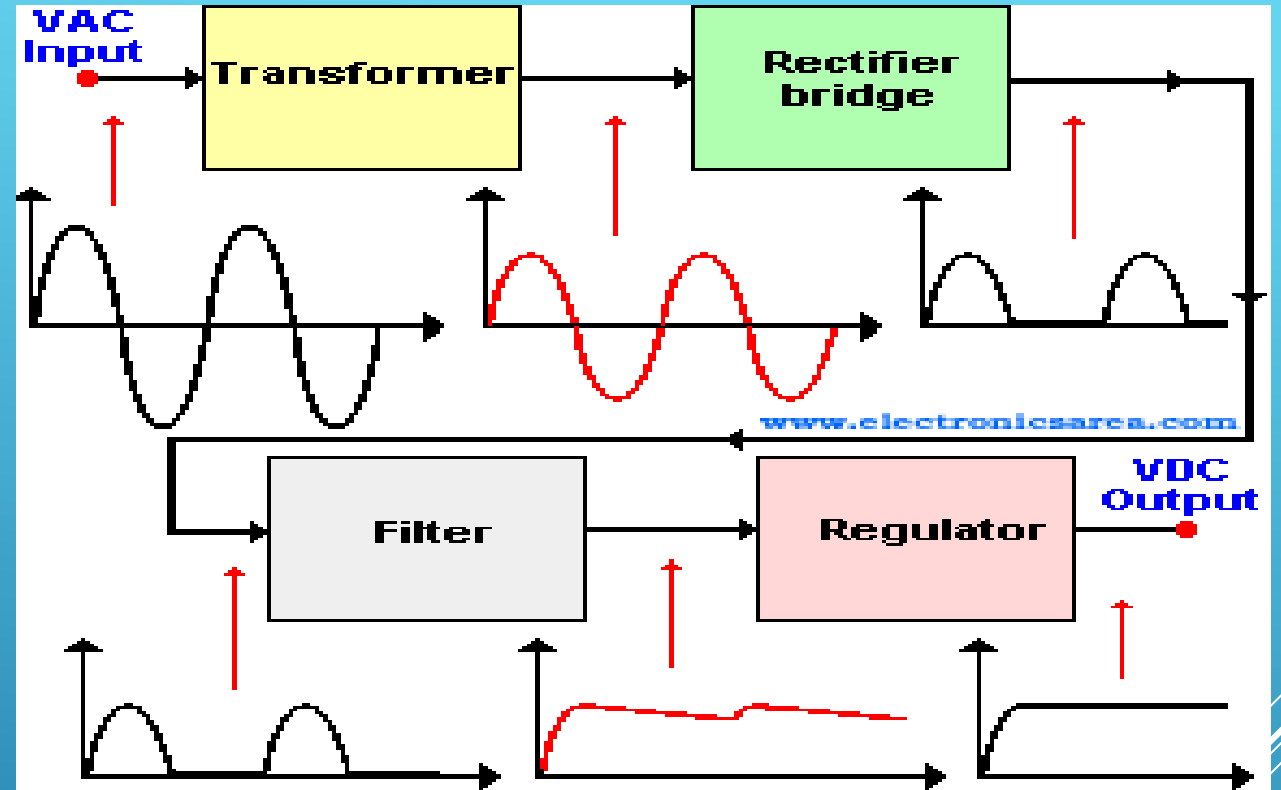
IC Voltage Regulators

A regulator Integrated Circuit (IC) unit contains the circuitry the reference source, comparator Amplifier , control device, and the overload protector – inside a single IC.

Power supply sample diagram:

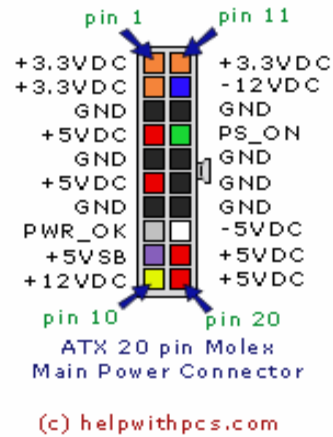
Color codes of power supply cables are as follow

1. Black wires – those are used to provide a ground for the current.
2. Yellow wires – this is color denotes +12volt
3. Red Wires – this color denotes +5volt
4. Blue Wires - -12volt
5. White Wires - -5volt
6. Orange Wires – 3.3volt
7. Green Wires- Control wire to check dc voltage
8. Purple Wires - +5V on standby mode

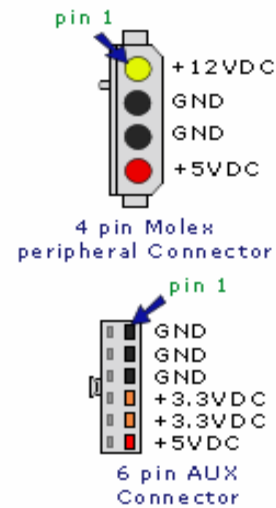


Power supply connectors

diagrams with pins facing forward



(c) helpwithpcs.com



Current/Latest power supplies

ATX / BTX motherboard power connector (20+4Pin)	Amount of connectors	Cable length	
	1	550mm	
EPS +12V Power Connector (4+4 Pin)	Amount of connectors	Cable length	
	1	550mm	
Peripheral connector (4 Pin)	Amount of connectors	Cable length	
	7	550mm	
Serial-ATA power connector	Amount of connectors	Cable length	
	8	550mm	
Floppy disc connector (4 Pin)	Amount of connectors	Cable length	
	1	550mm	
PCI-Express 6pin power connector	Amount of connectors	Cable length	
	2	550mm	
PCI-Express 6+2 pin power connector	Amount of connectors	Cable length	
	2	550mm	



Cables



ATX 24 PIN



4+4 PIN



PCI-E 6+2



SATA



**Molex
(PERIPHERAL)**



Floppy

Connectors

1

1

4

10

5

1

Thermal take smart SE 730W 80+ power supply

